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An Integrated Model for Optimizing Shale Gas Development Planning

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Abstract

Optimized development planning for large shale gas resource plays requires robust methods and models which incorporate both subsurface parameters and surface constraints. This paper presents a model which determines the optimal number of resource development blocks and facilities to achieve a defined annual natural gas production target.

The field development planning model is established to determine the optimal number of development blocks to achieve the defined production target while considering subsurface parameters and surface constraints. Subsurface parameters include type curves, shrinkage, products yields, and gas heat content. Surface parameters and constraints include the number of available blocks, well pads, surface infrastructure, and facility capacities. The model then determines the optimal number of primary facilities (i.e. gas separation plants) and secondary facilities (i.e. compression stations) while considering distance proximity. The sequential order of the development blocks is determined based on subsurface ranking of de-risking strategy.

Through the application of this model, development of large shale resource plays is optimized. Outputs include the number of resource blocks for development, number of well pads and associated wells, and necessary primary and secondary facilities, inclusive of location and operational year, along with the associated capacity utilization rate of each primary and secondary facility. With the number of wells determined, combined with the aforementioned infrastructure requirements, the production profile for each block and the aggregate development may be determined. Through this model, an enhanced shale development plan is determined to efficiently utilize necessary and available resources so as to maximize economic value for the developer.

This development model is independent from existing software applications. As an essential development planning tool, model outputs may be used as inputs into a petroleum economics model for further economic analysis to inform management decision making. Essential benefits derived from this model include enhanced development planning efficiency and improved strategic decision making. Use of this model may be further expanded to aid in cost reduction and reserves enhancement initiatives.

Background

Developing new unconventional fields requires substantial investment and a carefully structured development strategy. These projects typically span geologically complex and vast areas, which makes extraction both technically demanding and capital intensive. To navigate such challenges, operators adopt a phased development approach supported by detailed subsurface evaluation, resource modeling, and economic feasibility assessments. This stepwise process ensures that investments are optimized, risks are mitigated, and production is aligned with long-term strategic goals.

Unlike conventional reservoirs, unconventional developments involve tighter rock formations and lower permeability, requiring specialized technologies for commercial production. Horizontal drilling, multi-stage hydraulic fracturing, and advanced completion methods are commonly used to enhance reservoir contact and improve hydrocarbon recovery. These technologies also demand sustained capital deployment, making it critical to have clear visibility into the project's economic and technical performance across all phases of development.

Unconventional reservoirs, such as shale plays, tight sandstones, and carbonate formations often exhibit considerable geological heterogeneity. Properties like porosity, permeability, pressure, and saturation can vary significantly across even short lateral distances. This variability leads to greater subsurface uncertainty, which must be addressed through rigorous data integration, geostatistical modeling, and iterative evaluation workflows. Without these measures, there is a higher risk of over or underestimating the resource potential.

In addition, unconventional plays yield a wide range of hydrocarbon types, from dry gas to light oil and depending on factors like source rock maturity, depth, pressure regimes, and structural settings. For instance, the Eagle Ford shale in the U.S. has zones that produce dry gas, wet gas, or oil depending on their location within the basin. Similarly, the Wolfcamp formation in the Permian Basin shows lateral consistency but vertical heterogeneity in reservoir quality, requiring zone-specific development strategies.

These examples emphasize the importance of a tailored approach to field planning. A one size that fits all model rarely succeeds in unconventional development. As a result, modern field development relies heavily on integrated reservoir modeling, production and processing infrastructure optimization, production forecasting, and economic scenario analysis to support investment decision making. The ability to link subsurface understanding with that of the commercial outcomes has become a core competency in maximizing the value of unconventional assets.

Introduction

Efficient development planning of shale gas resources is essential for maximizing production potential and ensuring strong economic returns, particularly given the subsurface complexity and surface infrastructure constraints often associated with unconventional plays. Traditional planning methodologies frequently fall short, as they tend to overlook the integrated nature of geological, technical, and operational parameters that collectively influence field performance and project outcomes.

Unconventional shale reservoirs exhibit highly variable rock and fluid properties, demanding a more dynamic and data driven approach to resource development. In such contexts, aligning drilling schedules, facilities deployment, and production targets requires a robust decision-making framework that is capable of accommodating geological uncertainty and operational tradeoffs. To address these challenges, more sophisticated models are required, especially those that consider spatial heterogeneity, infrastructure constraints, and long-term economic optimization simultaneously.

This paper introduces a novel development planning model designed to improve resource block selection and facility placement strategies for large shale gas plays. The model incorporates key geological and surface parameters to determine the most effective sequence of development blocks, well pads, and associated infrastructure needed to meet a defined annual gas production target. By doing so, it offers a systematic and practical framework that enhances development efficiency, capital allocation, and risk management.

Ultimately, the model aims to support decision makers in achieving a more reliable and optimized development strategy, grounded in technical understanding and economic realism. It is particularly valuable for early phase planning and scenario evaluation in shale rich regions undergoing rapid development.

Field Overview

The synthetic reference field is characterized by four distinct hydrocarbon windows: Wet Gas Window (WGW), Gas Condensate Window (GCW), Volatile Oil Window (VOW), and High Condensate Gas Window (HCW). These classifications are based on the C7+ condensate fractions, as summarized in Table 1.0 below. Initial producing ratios (CGR, in stb/MMSCF) are also provided to describe production characteristics. The in-place volumes are considered sufficient to support long-term development.

Table 1—Properties of Petroleum Fluids1

Window	C7+ Fraction	Initial Producing Ratio (stb/MMSCF)
Dry Gas	C7+ < 0.5	CGR < 10
Wet Gas	0.5 < C7+< 4.0	10 < CGR < 70
Gas Condensate	4.0 < C7+< 12.9	70 < CGR < 310
Volatile Oil	12.9< C7+< 18.0	310< CGR < 530
Intermediate	18.0< C7+< 26.5	530 < CGR < 670
High CGR	C7+ > 26.5	670 < CGR

A strong understanding of how fluid properties vary across a field is essential for building reliable production forecasts. This includes knowing how condensate, natural gas liquids, and gas volumes, inclusive of initial production rates differ across various development areas. These differences directly impact the amount of gas and liquids that can be recovered, the design of processing facilities, and the economic value of each block. Proper classification into hydrocarbon windows, as outlined in the table above, provides the foundation for evaluating which zones are most viable and guiding the sequence of when they should be developed. By incorporating this knowledge into early stage planning, operators can make more informed decisions about drilling schedules, facility sizing, and commercial strategy.

Given the large size and technical complexity of unconventional developments (many of which require continuous drilling, high upfront spending, and phased infrastructure expansion) having a structured investment framework is critical. Without it, companies risk misallocating resources or facing delays due to poor coordination between subsurface and surface plans.

Fortunately, these types of developments allow for flexible and phased investment strategies. This means that companies can invest in stages, starting with lower risk, high return zones, and scaling up based upon learnings and results. From a business standpoint, this approach reduces capital exposure during early phases and supports improved financial control. By breaking the field into smaller, manageable development units linked to specific fluid windows, planners can better manage uncertainty and adjust as new data becomes available. Grouping these blocks into a larger development sequence further helps streamline operations, improve project timing, and maximize long term value across the full life of the asset.

Field Development Strategy

The development plan for the synthetic reference field includes drilling several thousand horizontal wells. Each well is designed to be hydraulically fractured and supported with artificial lift systems to help move fluids to the surface. The wells are connected to surface facilities using flowlines and remote gathering headers, which send the production to gas compression stations. From there, the compressed gas is delivered to a central gas processing plant.

At the gas plant, the production stream is separated and treated before it is sent to a fractionation facility, where valuable Natural Gas Liquids (NGLs) are extracted. These liquids, such as ethane, propane, and butane, are processed and sent to market through export pipelines. The remaining gas is cleaned and delivered as sales gas. Condensate is separated early in the process using a slug catcher and is also routed for further processing and export.

The development also includes supporting systems such as produced water handling and disposal systems, utility infrastructure such as electricity, water, and steam which are essential to maintain smooth and safe field operations. Figure 1 below shows a simplified diagram of the overall field development layout:

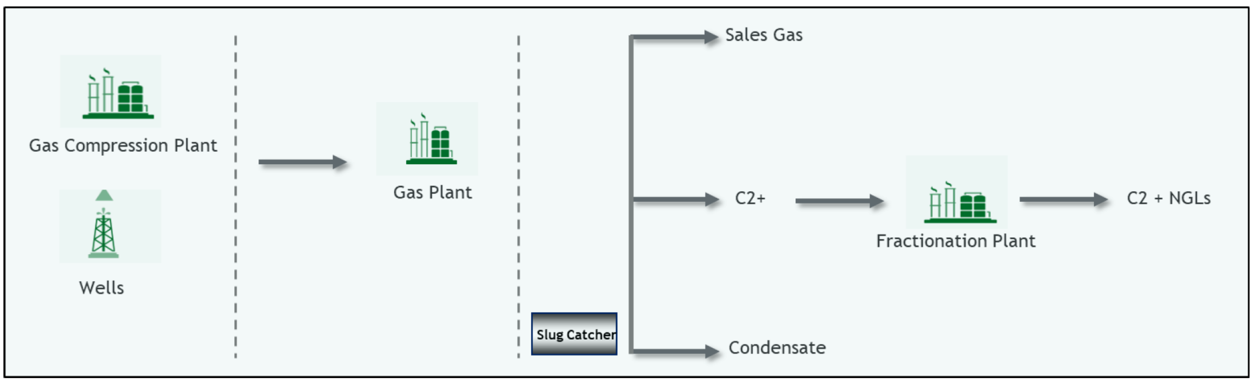

Figure 1—Conceptual Field Development Plan

The synthetic project is expected to last for around 20 years, starting with a 5-year initial development phase. During this period, drilling and infrastructure buildout will take place. Production is expected to begin toward the end of year 5, continuing through a 15-year plateau phase with steady output. The main product is natural gas, made up mostly of methane and small amounts of ethane, which represents the primary revenue source.

As with most unconventional developments, the processing stream also includes by-products such as NGLs, condensate, and small quantities of sulfur. These are part of the secondary production stream and offer additional commercial value. In this case, the NGLs are assumed to be fractionated and recovered at a high efficiency rate using a process such as Recycle Split Vapor (RSV). This method helps separate valuable liquids more effectively and supports better resource utilization.

Approach to Optimizing Shale Gas Development Planning

Planning the development of large shale gas fields can be challenging due to the differences in fluid types and limitations in surface infrastructure. To address this, a reliable model is needed that considers both the changes in fluid properties across the field and how easy it is to build and operate facilities. The model used in this study was designed to help in deciding how many development blocks are needed, where to place them, and what facilities are required in order to reach a specific gas production goal each year. At the same time, the model aims to improve infrastructure efficiency and increase the project's overall economic returns.

Model Structure and Key Inputs

The model begins by segmenting the field into four main fluid windows starting with Wet Gas Window (WGW), Gas Condensate Window (GCW), Volatile Oil Window (VOW), and High Condensate Window (HCW). These categories are based on the fluid composition in each area, particularly the ratio of condensate to gas. Each fluid window is then linked to a representative production curve that shows how fast production will decline over time and what total recovery can be expected. These curves are important because they help predict how each part of the field will perform and contribute to the success of the development.

To deal with differences in how easy it is to access different parts of the field, the model breaks the area into multiple development blocks. Each block is given an accessibility score which reflects how many wells are available and accessible, as well as reflecting how far the block is from key infrastructure facilities, how difficult it is to build, and how ready it is for operations. This score helps the model prioritize development blocks, typically starting with the those blocks ranked highest. This method supports a step by step development approach that reduces risk in the early phases. This approach further ensures that production starts from the most promising areas and that the number of wells drilled is enough to meet production goals while maximizing gas recovery.

As shown in Figure 2 below, the field is split into four different fluid windows with different inputs for each block.

Figure 2—Blocks distribution per fluid window

Another essential input is the shrinkage factor, derived from compositional sampling and laboratory analysis. It accounts for losses between the gas and surface deliverables, influencing volumetric forecasting and downstream processing assumptions.

Additionally, a fixed natural gas production target is defined to align the model outputs with long-term commercial objectives. The development drilling strategy is fundamentally one of ‘drill-to-fill’. This production strategy guides production pacing and block sequencing, ensuring delivery commitments are met without exceeding infrastructure capacity constraints.

Each development block is also linked to a predefined number of well pads and expected wells, based on surface constraints and available area. These inputs help determine spatial layout and facilitate more realistic infrastructure planning.

Model Outputs and Optimization Logic

Upon running the model, several types of results are generated. The most important output is the projected gas production rate, which is calculated for each individual and subsequently aggregated across the entire field. This gas production forecast helps to show how much gas is expected over time and supports long term planning.

In addition to natural gas, the model also estimates volumes of by-products such as ethane, methane, propane, butane, pentane, condensate, and sulfur. These are called secondary outputs, and they are calculated using fluid composition data collected earlier during reservoir analysis. These outputs help in planning for processing capacity, product sales, and infrastructure needs. An example of the primary output which is the projected gas rate over 20 years, is shown in Figure 3 below.

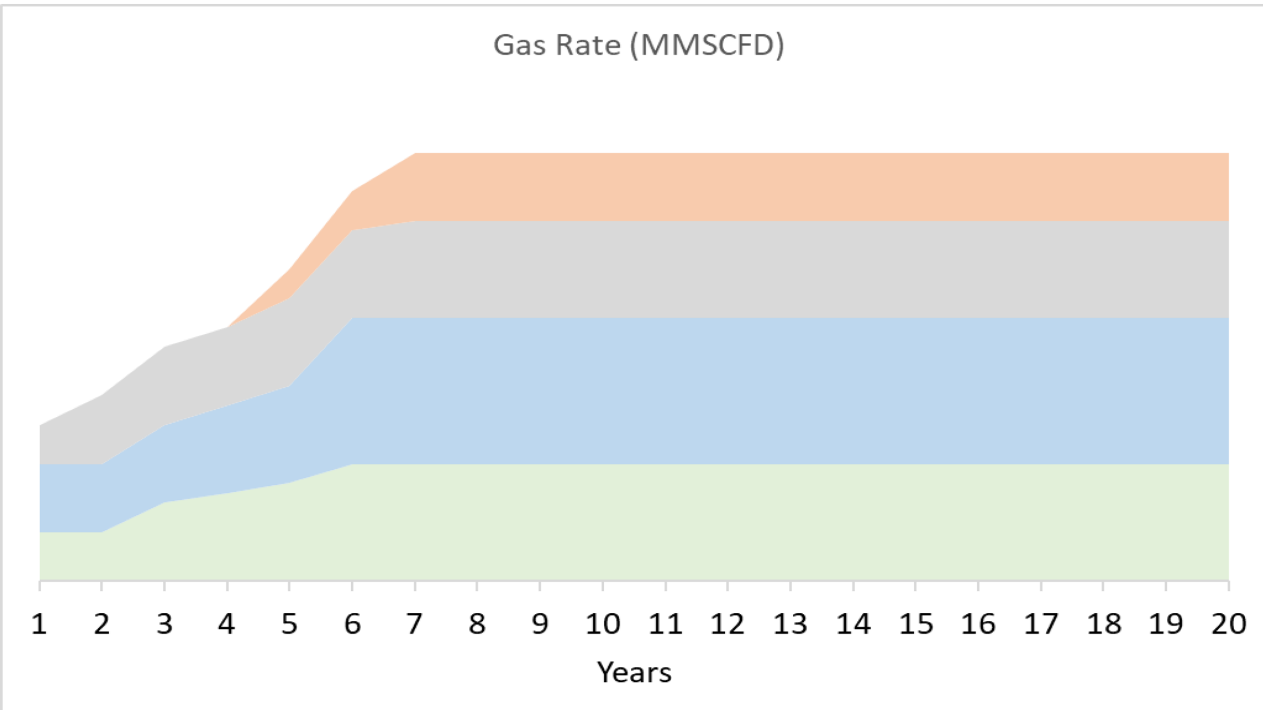

Figure 3—Gas rate as a primary output

To make the development process more efficient, the model automatically attributes production across different blocks in the field. It starts by choosing blocks that are easier to reach and expected to be more productive. This approach ensures that early stage investments are made in areas where the risk is lower and returns are higher.

The model also checks how production volumes match with the capacity of surface facilities, such as gas compression plants and fractionation units. It applies constraints to make sure no facility handles more gas than it is designed for. This avoids overloading any part of the system and helps identify when expansions or upgrades will be needed in the future, also known as debottlenecking.

The final setup of the model creates an optimized development plan, showing the best order for drilling blocks, placing wells, and building required facilities. This plan aims to make the best use of subsurface potential while staying within surface limitations like land area, access, and infrastructure capacity.

By using this method, operators can run different development scenarios, test how each one performs, and make better decisions on where and when to invest. This makes the model a powerful tool for strategic planning and helps improve both technical outcomes and economic performance.

Figure 4 below summarizes the key inputs and outputs of the model. Inputs include things like gas production targets, shrinkage factors, type curves, and available wells. Outputs include total gas volume, well counts, block selection, by-product estimates, and a final field layout map.

Inputs						
Sales gas target	Shrinkages	Type curves	Well inventory	Block Ranking	GCP capacity	Constraints

Outputs						
GCP locations, count & kickoff	GCP utilization	Total well count	Sales gas volume	Block distribution	By-products profiles	Field map view

Figure 4—Model Structure

Conclusion

Successfully developing a gas field requires more than just accurate data or good technical models. It requires a complete and connected approach that links produciton targets with technical decisions and robust operational plans. To achieve this, companies must align well inventory, surface facility limitations, and production targets using a structured and flexible planning method.

The model presented in this paper supports this kind of integrated development. It allows different types of data starting from geological inputs through to infrastructure considerations and constraints to be processed and turned into actionable plans. One of the biggest advantages of this approach is the ability to simplify complex data into clear outputs, making it easier for decision makers to plan effectively. This improves forecasting accuracy, helps avoid delays from bottlenecks, and supports better timing in project execution.

Another important benefit of this method is how it encourages teamwork between different technical groups, such as subsurface engineers, production planners, and facilities teams. By working with the same model and using the same inputs, teams can communicate more easily and make faster, more consistent decisions. This is especially useful in large projects with many moving parts, where misalignment can cause delays or reduce efficiency. As unconventional gas developments grow in size and complexity, having a data-driven and flexible planning tool becomes even more critical. The ability to test different development scenarios, understand trade offs, and make adjustments in real time helps companies stay on track with short term production goals while also protecting long term performance.

Ultimately, using a well structured and holistic development model improves operational readiness, reduces uncertainty, and increases the overall value of the project. It helps organizations move from reactive problem solving to proactive planning, enabling them to maximize returns over the full life of the asset.

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